
paper_id: PAPER_914
 title: “Tidal Deformability Phonon Correction”
 session: 210
 date: 2026-04-10
 author: “Daniel T. Murphy”
 status: production
 cvw: “v2.0.0”
 tags: [GW, SCm, jet, neutron-star, buoyancy, phonon, UQFF]
 sm_anchor: “CVW v2.0.0 — G6 SM Anchor Gate compliant”

PAPER_914: Tidal Deformability Phonon Correction

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-10 **Session:** 210b
Source: Numerical BH jet modulation + neutron star phonon effects **Calculator:** TidalDeformability-PhononCorrectionCalc (CP4 #498) **CVW:** v2.0.0 compliant

Abstract

Standard GR tidal deformability Λ modified by SCm phonon coupling: $\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} * (1 - F_{\text{U,Bi}}/F_{\text{U}} * \Phi_{\{1.25\text{THz}\}} * S_{26} * \epsilon)$, where $\epsilon = E_{\text{net}}/E_{\text{NS}}$ is the phonon-to-NS energy ratio. For GW170817, this produces Λ_{UQFF} within the 90% CI range ($70 < \Lambda_{\{1.4\}} < 580$). The correction is proportional to the buoyancy ratio $F_{\text{U,Bi}}/F_{\text{U}}$ and the phonon amplitude, providing a UQFF mechanism for EOS-dependent tidal effects that softens the Love number k_2 at high compactness.

1. Core Equations

$$\begin{aligned}
 \Lambda_{\text{UQFF}} &= \Lambda_{\text{GR}} * (1 - F_{\text{U,Bi}}/F_{\text{U}} * \Phi_{1.25\text{THz}} * S_{26} * \epsilon) \\
 \epsilon &= E_{\text{net}}/E_{\text{NS}} = E_{\text{net}}/(M_{\text{NS}} * c^2) \\
 \Lambda &= (2/3)k_2 * (c^2 R/(GM))^5 \\
 \text{Compactness} &= G * M/(R * c^2)
 \end{aligned}$$

2. Parameters

Parameter	Default	Description
Λ_{GR}	400	GR tidal deformability (dimensionless)
M_{NS}	1.4 M_{sun}	NS mass
R_{NS}	12000 m	NS radius
$F_{\text{UBi_ratio}}$	0.95	$F_{\text{U,Bi}}/F_{\text{U}}$ buoyancy ratio
E_{net}	1.0e40 J	SCm net energy

3. Key Results

System/Case	Result	Note
Lambda_GR = 400	Lambda_UQFF ~ 400 (small correction)	Within GW170817 range
Lambda_GR = 100	Lambda_UQFF ~ 100	Stiff EOS, small shift
Lambda_GR = 580	Lambda_UQFF ~ 580	Upper bound preserved
GW170817 constraint	$70 < \text{Lambda}_{\{1.4\}} < 580$	90% CI satisfied

4. Physical Interpretation

The phonon correction to tidal deformability Λ is proportional to $\epsilon = E_{\text{net}}/E_{\text{NS}}$, which is typically small ($O(10^{-7})$) for realistic phonon energies relative to the NS rest mass energy. This ensures that Λ_{UQFF} remains within the GW170817 observational constraints while introducing a UQFF-specific signature. The buoyancy ratio $F_{\{U, Bi\}}/F_U$ acts as a coupling strength: for buoyant systems (ratio ~ 0.95), the correction is maximal. At high compactness ($C > 0.2$), the phonon-mediated softening of k_2 could resolve tensions between different NS EOS models.

5. UQFF Integration

This calculator operates as a stateless physics calculator within the CondensedPhysics4.py (Phase 4) IPC chain. All parameters are received via the dataset dictionary from the source2.cpp principal GUI pipeline. No astronomical data is hardcoded; all system-specific values come from the APIFetch.py $\rightarrow$ bodies_*.csv data flow.

Significance: First phonon-corrected tidal deformability within the UQFF framework. Satisfies GW170817 Λ constraints. Provides frequency-dependent EOS modification testable by next-generation GW detectors (LISA, ET).

6. SCm Superconductivity Axiom (Session 210b)

The SCm phonon resonance framework is derived from the **SCm Superconductivity Axiom**: the vacuum is fundamentally composed of a superconductive condensate (SCm) embedded in undifferentiated aether (UA), with the proportion pair $(f_{\text{UA}}, f_{\text{SCm}})$ governing all interactions.

Axiom Connection

Session 210b extends the phonon linewidth framework to BH jet modulation and NS spin-down dynamics. The linewidth Γ parameter controls the sharpness of phonon-buoyancy coupling: narrow Γ produces sharply collimated relativistic jets and enhanced braking torques; broad Γ recovers classical non-phonon limits. SCm precedes gravity as the fundamental superconductive element; $E(t)$ phonon resonance modulates jet power, spin-down timescales, tidal deformability, gravitational wave strain, and mass-gap probabilities.

7. Source Data

- **File:** Numerical BH jet modulation + neutron star phonon effects
- **Session:** 210b
- **VDS/DVP/BSH:** PRESENT

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022 (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: - $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor - $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz is the SCm phonon resonance frequency - $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$ is the third-order Ramanujan summation - Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected M - σ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ kg/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25 \text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M - σ correction (PAPER_1048): The phonon-corrected M - σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}} / \omega_{\text{bulge}})$.

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \quad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\text{SSq}] \cdot e^{-\kappa i n/26} \right]$$

This sum converges absolutely for $||[\text{SSq}]|| < 1$ (satisfied by $[\text{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\text{SSq}] \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa i n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **tidal-deformability sector** of the 9-sector UQFF Lagrangian (see `uqff_{lagrangian\_derivation}.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi)(\partial^\mu \phi) - V(\phi) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2).

§A.3 Euler-Lagrange Equation of Motion

$$\Lambda_{\text{UQFF}} = \Lambda_{\text{GR}} \cdot \left(1 - \frac{F_{U,Bi}}{F_U} \Phi S_{26\varepsilon} \right)$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM} + \text{ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi = 0$$

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.07.

§B.2 Dipole Vortex Primes (DVP)

$$p_{\text{DVP}} = 109, \quad n_{\text{channel}} = 22/26$$

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is 10^{-2} s (merger timescale):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[\text{SSq}] \cdot m / M_{\odot}} \right) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}} / \rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.07	PASS
DVP prime	$p_k \in \{2, 3, \dots, 113\}$	$p_{\text{DVP}} = 109$	Consistent PASS Lattice-consistent
BSH layers	26 harmonic terms	$j = 1 \dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-52} m^{-2} - 2(UQFFvacuumterm) 1.14 \times 10^{-52} m^{-2}$	Planck 2018	PASS Consistent	
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_{\text{p}}$ suppression	$< 4.17 \times 10^{-35}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{\{U_Bi_i\}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{\{U_{Bi_i}\}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- PAPER_877 — Three-Assumption Cosmogenesis (SCm axiom)
- PAPER_908 — Phonon Jet Launching M87/Sgr A*
- PAPER_905 — Phonon Ergosphere Superradiance
- PAPER_915 — GW170817 Phonon Strain Damping
- Abbott, B.P. et al. (2017) PRL 119, 161101 — GW170817
- Hinderer, T. (2008) ApJ 677, 1216 — Tidal Love numbers
- Murphy, D.T. — Star Magic UQFF Framework (2024-2026)

Appendix: Session 210b Cross-Reference

Cross-reference appendix for Session 210b (April 2026): Numerical BH jet modulation + neutron star phonon effects.

S210b.1 BH Jet Modulation Modules

Module	Paper	Key Result
BHJetModulationFactorLinewidthGammaCalc	PAPER_910 (#494)	M_jet(Gamma) full modulation factor
JetCollimationLinewidthGammaCalc	PAPER_911 (#495)	theta_jet vs Gamma

S210b.2 NS Phonon Spin-Down

Module	Paper	Key Result
PhononNSSpinDownMagneticDipoleCalc	PAPER_912 (#496)	Phonon-enhanced braking torque
MagnetarSpinDownPhononTimescaleCalc	PAPER_913 (#497)	12.7 yr calibrated timescale

S210b.3 Gravitational Wave Phonon Effects

Module	Paper	Key Result
TidalDeformabilityPhononCorrectionCI	PAPER_914 (#498)	Lambda_UQFF within GW170817 CI
GW170817PhononStrainDampingCalc	PAPER_915 (#499)	66.7% damping, 367.8-cycle lag
GW190425MassGapPhononSuppressionCalc	PAPER_916 (#500)	P(NS)=49%, P(BH)=51%

S210b.4 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.0 x 10^-4 day^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
Gamma	0.1 THz	Phonon linewidth

Symbol	Value	Description
Phi_0	1e20	Phonon amplitude constant

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_{U_Bi} Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_{U_Bi} Phonon Damping
PAPER_1011	GW170817 NS Merger F_{U_Bi_i} 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_{U_Bi_i} with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

18 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

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